For Examiner's Use

7 **(a)** A radiation detector is placed close to a radioactive source. The detector does not surround the source.
Radiation is emitted in all directions and, as a result, the activity of the source and the measured count rate are different.

Suggest **two** other reasons why the activity and the measured count rate may be different.

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..... [2]

(b) The variation with time *t* of the measured count rate in **(a)** is shown in Fig. 7.1.

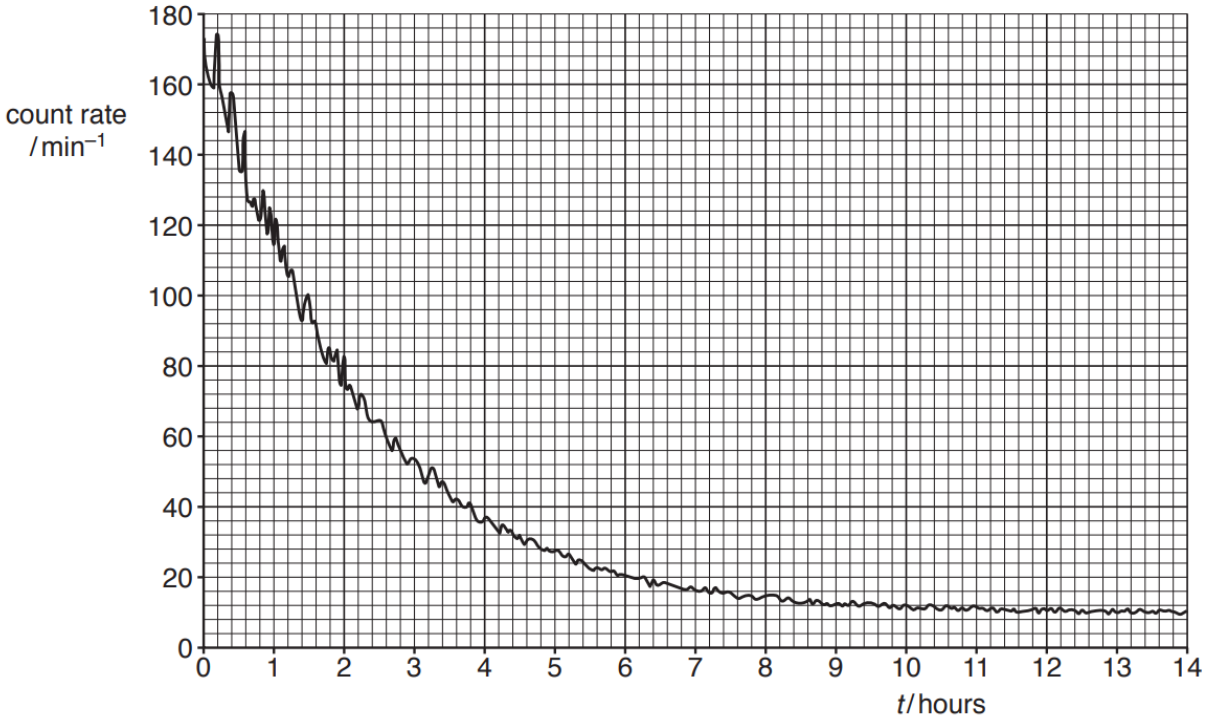

Fig. 7.1

(i) State the feature of Fig. 7.1 that indicates the random nature of radioactive decay.

.....

..... [1]

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(ii) Use Fig. 7.1 to determine the half-life of the radioactive isotope in the source.

half life = ... s [3]

(c) The readings in (b) were obtained at room temperature.

A second sample of this isotope is heated to a temperature of 500°C.

The initial count rate at time t = 0 is the same as that in **(b)**.

The variation with time t of the measured count rate from the heated source is determined.

State, with a reason, the difference, if any, in the half-life.

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..... [1]